

FINAL YEAR PROJECT – COMPLETE & LOCKED SPECIFICATION

◆ Project Title

EduGenie – AI-Powered Syllabus-Focused Study Assistant for MCA Students

◆ Project Overview

EduGenie is a **full-stack AI-powered web application** designed to help MCA students study more effectively by generating **exam-oriented notes**, answering **subject-specific academic doubts**, explaining **programming code**, and providing **resume and project guidance**.

Unlike generic AI chatbots, EduGenie is **syllabus-aware** and focuses on **structured, curriculum-aligned learning**, making it suitable for academic use as well as interview preparation.

The system uses a **secure API-based architecture**, integrates **AI through prompt engineering**, and stores user learning history in a **relational database** for better organization and retrieval.

◆ Objectives

- Help students generate **clear, structured, exam-ready notes**
 - Provide **subject-wise academic explanations**
 - Assist with **coding logic and debugging**
 - Offer **resume and project guidance for freshers**
 - Demonstrate **real-world full stack development skills**
 - Build a **job-ready, interview-defensible project**
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◆ Target Users

- MCA students
 - Computer science students preparing for exams and placements
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◆ Core Features

Student/User Features

1. User Authentication

- Signup & Login
- JWT-based authentication
- Secure password hashing

2. Subject & Syllabus Selection

- Subject-wise organization (OS, CN, DBMS, OOP, etc.)
- Topic-based learning

3. AI Notes Generator

- Generates structured, exam-oriented notes
- Simple language explanations
- Cached notes to reduce AI usage

4. Doubt Solving / AI Chat

- Subject-specific academic Q&A
- Context-aware responses
- Chat history saved for revision

5. Programming Code Explainer

- Explains logic step-by-step
- Identifies errors
- Supports languages like Python, C, Java
- Focuses on explanation, not blind code generation

6. Resume & Project Guidance

- Role-based suggestions (Frontend / Backend / Fresher)
- Improves resumes using controlled AI prompts
- Provides project improvement ideas

7. History Management

- Saved notes
- Saved chats
- User-specific data storage

⚙️ System / Admin Features

- REST API-based backend architecture
- Secure JWT authentication
- Prompt-based AI response system
- Relational database storage
- Error handling & validation
- Rate limiting and usage control
- Cloud-ready deployment setup

◆ Technology Stack (FINAL)

Frontend

- React
- Basic CSS / Tailwind (optional)
- Hosted on **Vercel (Free Tier)**

Backend

- **Python + Flask**
- RESTful APIs
- JWT Authentication

Database

- **MySQL**
- Structured relational schema
- Normalized tables with foreign keys

AI Integration

- **Ollama (Local AI Models)**
(e.g., LLaMA / Mistral)
- Prompt engineering for:
 - Notes

- Explanations
- Code logic
- Resume guidance

Security

- JWT for authentication
 - bcrypt for password hashing
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◆ System Architecture (High-Level)

Frontend (React)

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| REST APIs (JWT)
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Backend (Flask)

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| Prompt Templates
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AI Engine (Ollama)

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MySQL Database

◆ Database Design (High-Level Tables)

- users
- subjects
- topics
- notes
- chat_history
- resume_feedback

Key concepts demonstrated:

- Primary keys
 - Foreign keys
 - One-to-many relationships
 - Normalization
 - SQL queries & joins
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◆ **Project Scope Control (IMPORTANT)**

Included

- Structured academic assistance
- Controlled AI usage
- Exam-focused outputs
- Clean UI and backend logic

Not Included (Intentionally)

- Unlimited AI usage
 - AI model training from scratch
 - Production-scale deployment
 - Replacement of teachers or textbooks
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◆ **Limitations**

- AI responses depend on prompt quality
 - Local AI may be slower than cloud-based systems
 - Not designed for large-scale public usage
 - AI may occasionally produce incorrect responses
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◆ **Deployment Strategy (Free Tier)**

- Frontend → Vercel
- Backend → Render / Local demo

- Database → Local or Cloud MySQL
 - AI → Local (Ollama)
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◆ Role Distribution (Team Project)

Backend & AI (You)

- Flask APIs
- Authentication
- Database schema
- AI logic & prompts
- Deployment
- Architecture explanation

Frontend (Teammate)

- UI/UX
 - Dashboard
 - Chat interface
 - Notes display
 - History pages
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◆ Project Outcome

- A complete **full-stack AI-integrated system**
 - Strong **final year project submission**
 - Resume-ready project for **developer job roles**
 - Clear talking points for **interviews and vivas**
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◆ One-Line Interview Pitch

“EduGenie is a syllabus-focused AI study assistant for MCA students that generates exam-ready notes, explains concepts and code, and stores learning history using a secure full-stack architecture.”